

# Stack Implementation Using pointer

```
#include <stdio.h>
#include <conio.h>

struct stack
{
    int no;
    struct stack *next;
} *top = NULL;

typedef struct stack st;

void push();
int pop();
void display();
void main()
{
    int ch;
    char ch1;
    int item;
    do
    {
        clrscr();
        printf("\n 1 -> push");
        printf("\n 2 -> pop");
        printf("\n 3 -> Display");
        printf("\n 4 -> Exit");
        printf("\n %d", ch);
        switch (ch)
    }
}
```



case 1: push();

break;

Case 2: printf("the deleted item is %d", pop());

break;

Case 3: display();

break;

Case 4: exit();

default: printf("wrong choice");

printf("Do you want to continue Y/N");

scanf("%c", &ch1);

while (ch1 == 'y' || ch1 == 'Y');

void push()

{

st \*node;

node = (st \*) malloc(sizeof(st));

printf("Enter the number");

scanf("%d", &node->no);

node->next = top;

top = node;

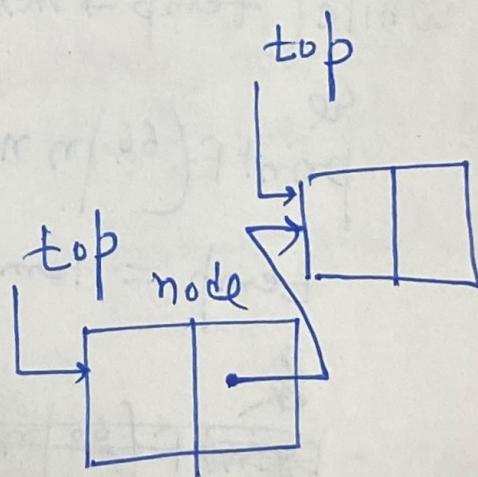
}

int pop()

{

st \*temp;

temp = top;





```
if (temp == NULL)
```

```
{  
    printf("\n stack is empty");  
    exit();  
}
```

```
{
```

```
    else
```

```
        top = top → next;
```

```
        Free(temp);
```

```
}
```

```
    return(temp → no);
```

```
}
```

```
void display()
```

```
{
```

```
    st *temp;
```

```
    temp = top;
```

```
    while (temp → next != NULL)
```

```
    {  
        printf("\n number is %d", temp → no);
```

```
        temp = temp → next;
```

```
    }  
printf("\n the number is %d", temp → no);
```

```
}
```